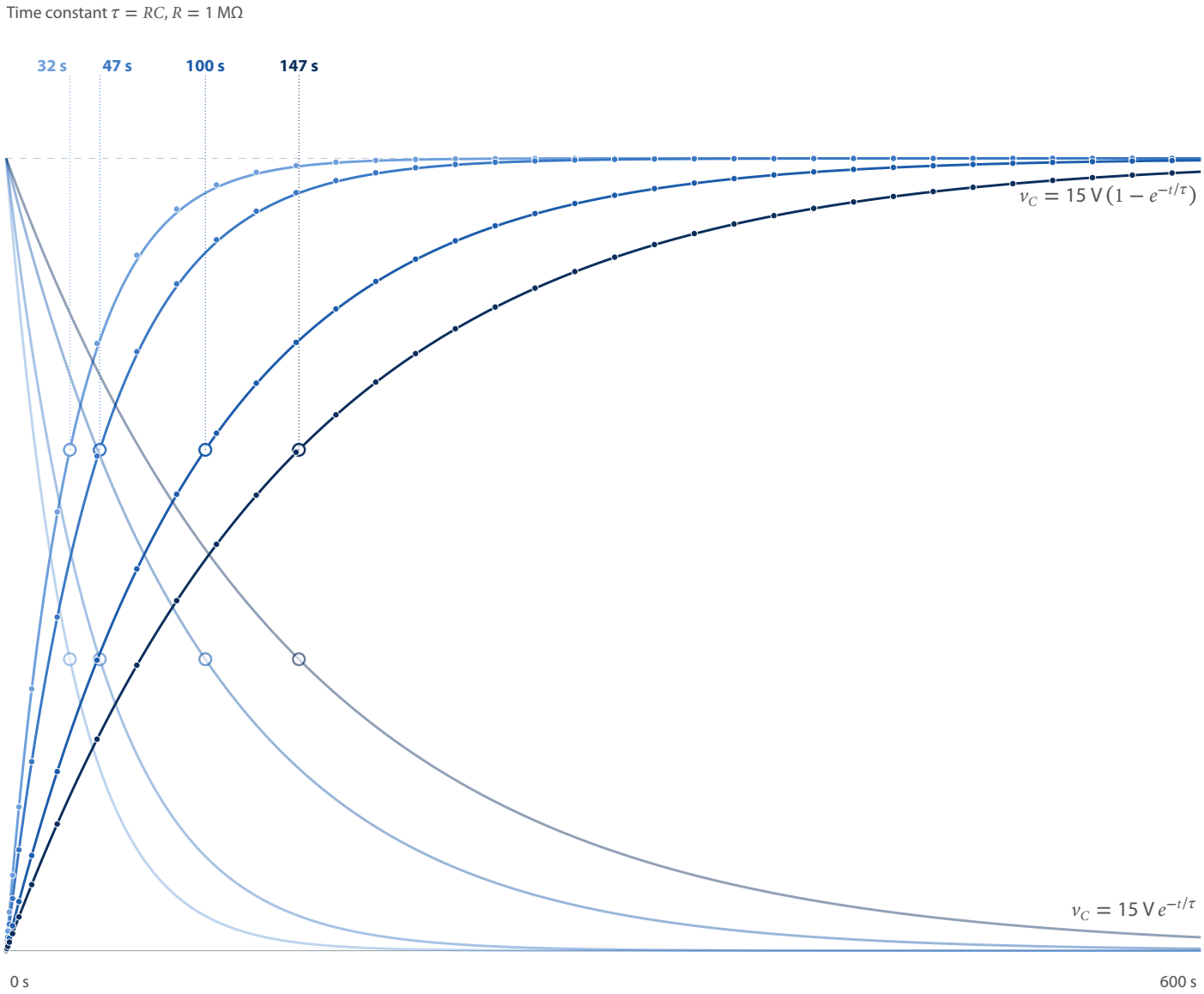
Leonard H Phelan IV

RC Transient Response



Part I: Prelab Calculations

All four circuits charge through the same 1 MΩ from the same 15 V, so the time constant is set by the capacitance alone: 47 s, 100 s, 147 s and 32 s. Every circuit reaches 9.48 V after 1τ charging and falls to 5.52 V after 1τ discharging, and the current starts at 15 µA in all four.

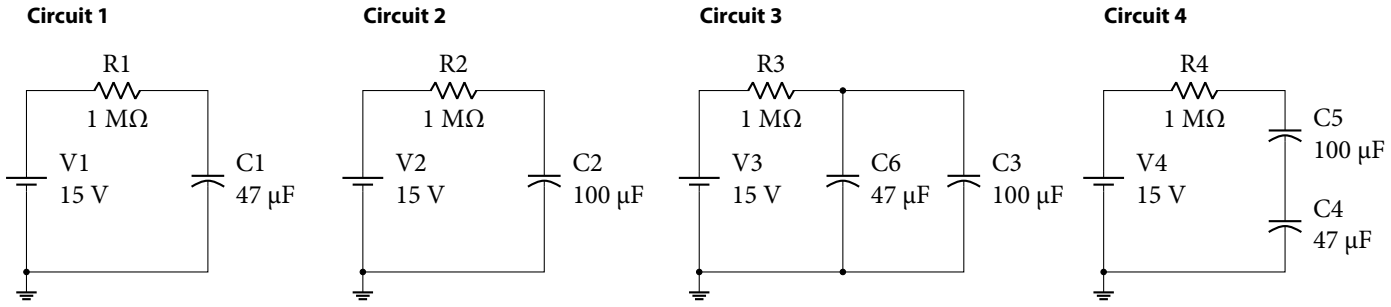


Figure 1. The four RC circuits from the Part I table, redrawn from the handout.

1.1 Equivalent capacitance

Capacitors in parallel add, so Circuit 3 is 147 µF:

$$C_{\mathrm{EQ}} = 47\ \mu\mathrm{F} + 100\ \mu\mathrm{F} = 147\ \mu\mathrm{F} \tag{1}$$

Capacitors in series combine the way parallel resistors do, so Circuit 4 is smaller than either capacitor:

$$C_{\mathrm{EQ}} = \frac{(47\ \mu\mathrm{F})(100\ \mu\mathrm{F})}{47\ \mu\mathrm{F} + 100\ \mu\mathrm{F}} = 31.97\ \mu\mathrm{F} \approx 32\ \mu\mathrm{F} \tag{2}$$

1.2 Time constants and expected values

With $R = 1$ MΩ, each microfarad adds one second,[1] so τ in seconds equals the capacitance in µF:

$$\tau = RC = (1\ \mathrm{M\Omega})(47\ \mu\mathrm{F}) = 47\ \mathrm{s} \tag{3}$$

While charging, the capacitor voltage climbs toward V_S and gets 63.2 % of the way there in one time constant:

$$v_C(t) = V_S\left(1 - e^{-t/\tau}\right) \qquad v_C(\tau) = 0.632 \cdot 15\ \mathrm{V} = 9.48\ \mathrm{V} \tag{4}$$

While discharging, it falls from its starting voltage and has 36.8 % left after one time constant:

$$v_C(t) = V_0\, e^{-t/\tau} \qquad v_C(\tau) = 0.368 \cdot 15\ \mathrm{V} = 5.52\ \mathrm{V} \tag{5}$$

At $t = 0$ all 15 V sits across the 1 MΩ: when charging because the empty capacitor acts as a short, and when discharging because the full capacitor is the source. That is the maximum current, and it decays with the same τ either way:

$$I_{C,\max} = \frac{V_S}{R} = \frac{15\ \mathrm{V}}{1\ \mathrm{M\Omega}} = 15\ \mu\mathrm{A} \tag{6}$$

$$i_C(\tau) = 0.368 \cdot 15\ \mu\mathrm{A} = 5.52\ \mu\mathrm{A} \tag{7}$$

[1] $1\ \mathrm{M\Omega} \times 1\ \mu\mathrm{F} = 10^{6}\ \Omega \times 10^{-6}\ \mathrm{F} = 1\ \mathrm{s}$.

Time	Charging	Discharging
1τ	9.48 V	5.52 V
2τ	12.97 V	2.03 V
3τ	14.25 V	0.75 V
4τ	14.73 V	0.27 V
5τ	14.90 V	0.10 V

Table 1. Capacitor voltage after each time constant, from 15 V. After 5τ it is within 0.7 % of the end value, the usual point to call it fully charged or discharged.

None of the voltages or currents depend on the capacitance; only the time it takes to reach them does (Table 2).

	Circuit 1	Circuit 2	Circuit 3	Circuit 4
Configuration	1 MΩ + 47 µF	1 MΩ + 100 µF	1 MΩ + (47 µF ‖ 100 µF)	1 MΩ + (47 µF in series with 100 µF)
Capacitance	47 µF	100 µF	147 µF	31.97 µF
$\tau = R \cdot C$	47 s	100 s	147 s	31.97 s
Source	15 V	15 V	15 V	15 V
V_C after 1τ, charging	9.48 V	9.48 V	9.48 V	9.48 V
V_C after 1τ, discharging	5.52 V	5.52 V	5.52 V	5.52 V
Maximum I_C	15 µA	15 µA	15 µA	15 µA
I_C after 1τ	5.52 µA	5.52 µA	5.52 µA	5.52 µA
V_C after 5τ, charging	14.90 V	14.90 V	14.90 V	14.90 V

Table 2. Part I prelab values. The last two rows are not in the handout's table; Parts II and III use them.

Part II: RC Transient Response (Charging)

In Multisim each capacitor reaches 63.2 % of 15 V at its own time constant. Circuits 1–3 match the calculated 9.48 V and 5.52 µA within 0.4 %; Circuit 4 reads 1.7 % low on voltage and 2.8 % high on current.

2.1 Multisim

Circuit 1 was built from the 15 V source, 1 MΩ and 47 µF, with an AV probe reading VDC and IDC and its arrow pointing into the capacitor. A Transient analysis ran to TSTOP = 1000 s with the initial conditions set to User-Defined.[2]

The current trace was moved to a right-hand axis (Figure 4). Circuits 2–4 were then built beside it with each probe renamed to its time constant (Figure 3), and the analysis was re-run plotting the four voltages (Figure 5) and then the four currents (Figure 6). Instead of reading values off the Grapher's cursors, each plot was exported to Excel, and the values at 1τ and 5τ were interpolated between the two nearest exported points.

[2] Multisim chose the spacing of the saved points itself: the export doubles from 0.2 s up to 25.6 s, then holds 20 s apart.

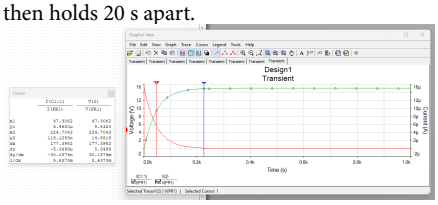


Figure 2. Multisim's Grapher with Circuit 1's two traces, before the export. Cursor 1 at 47.3 s *reads* 9.532 V *and* 5.468 µA.

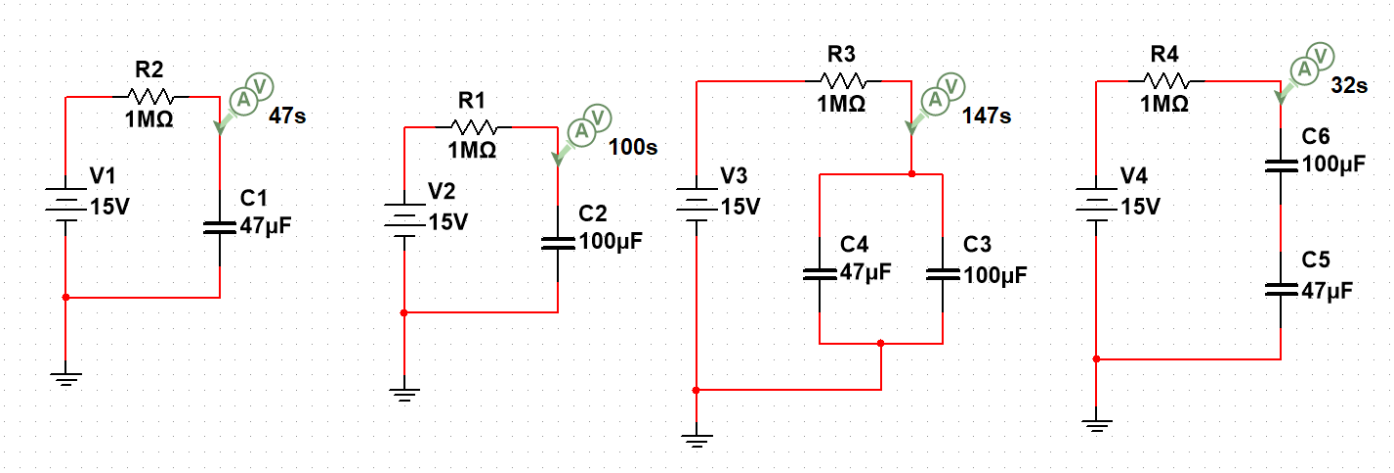


Figure 3. Part II circuits in Multisim. Each AV probe is named for its circuit's time constant and points into the capacitor.

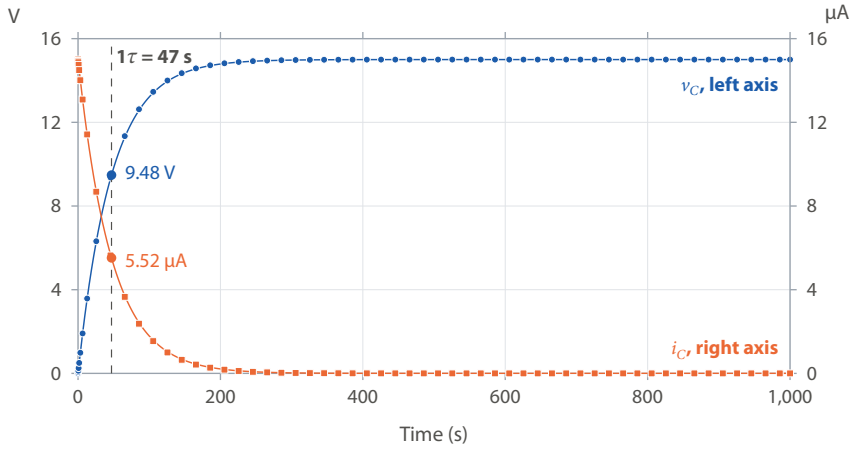


Figure 4. Circuit 1 charging: capacitor voltage and current, with $1\tau = 47$ s marked. Lines are calculated values; marks are the exported points.

2.2 Results

For Circuit 1 alone, the exported data at 47 s reads 9.502 V and 5.498 µA, within 0.4 % of the calculated 9.48 V and 5.52 µA. Circuits 2 and 3 agree as closely; Circuit 4 is the outlier (Table 3).

Circuit	τ (s)	V_C at 1τ (V)	Error (%)	I_C at 1τ (µA)	Error (%)	V_C at 5τ (V)
1	47	9.502	+0.23	5.498	−0.39	14.904
2	100	9.474	−0.07	5.526	+0.12	14.900
3	147	9.486	+0.06	5.514	−0.11	14.899
4	31.97	9.323	−1.66	5.677	+2.85	14.909

Table 3. Multisim values at 1τ and 5τ while charging, from the exported data. Errors are against the calculated 9.48 V and 5.52 µA; all four should read 14.90 V at 5τ.

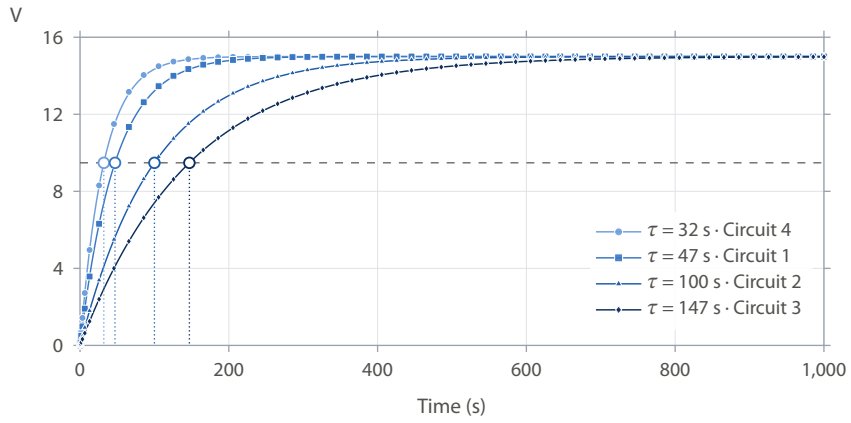


Figure 5. Capacitor voltage while charging, all four circuits. Each crosses 9.48 V *at its own time constant.*

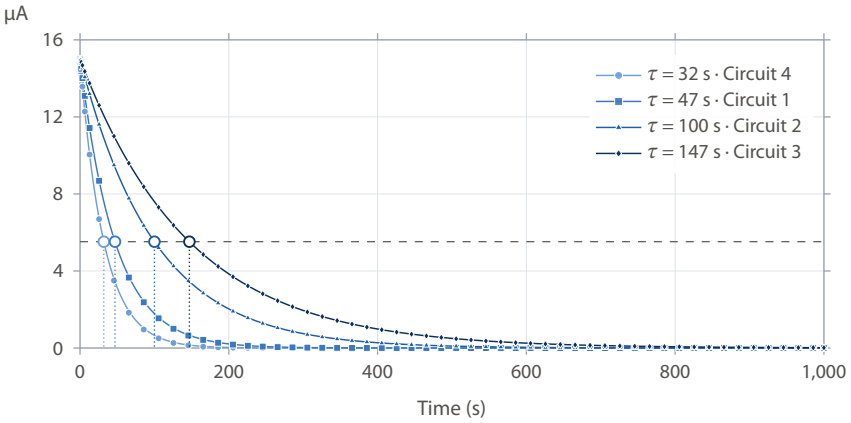


Figure 6. Capacitor current while charging. Every circuit starts at 15 μA *and falls to* 5.52 μA *at its own time constant.*

Circuit 4 charged fastest and Circuit 3 slowest, the same order as their time constants: 32, 47, 100, then 147 s (Figure 5). More capacitance takes longer to fill through the same 1 MΩ. All four reached the same voltage after 5τ, 14.90 V within 0.01 V; they just took between 160 s and 735 s to get there.

The currents rank the same way (Figure 6). They have to, because the current is set by the voltage still left across the resistor, $(V_S - v_C)/R$. Once the capacitor is 63.2 % charged, the current is down to 36.8 % of its start, and every circuit starts at 15 μA because R is the same. In each row of Table 3 the two readings add back to the source: $9.502\ \text{V} + (5.498\ \mu\text{A})(1\ \text{M}\Omega) = 15.000\ \text{V}$.

Circuit 4's error is in the export's point spacing, not the circuit. With TSTOP = 1000 s, Multisim saved points 20 s apart after the first 25 s (25.6 s, 45.6 s, 65.6 s and so on), so the export has no point at Circuit 4's 1τ of 32 s. Its value there comes from a straight line between the points at 25.6 s and 45.6 s, and a 20 s gap is almost two-thirds of its τ, so the line cuts under the curve (Figure 7). The saved points themselves are within 0.1 V of the formula. For Circuits 1–3 the same gap is a smaller fraction of τ, and the error stays under 0.4 %.

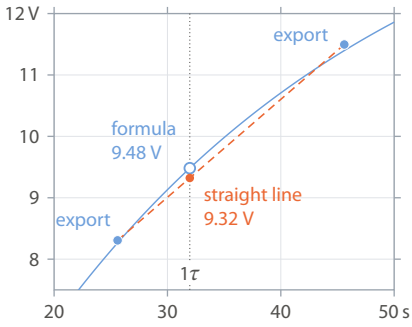


Figure 7. Circuit 4 near its 1τ. *The export's nearest points are* 20 s *apart, and the straight line between them passes* 0.16 V *under the curve at* 31.97 s.

Part III: RC Transient Response (Discharge)

With the sources removed and every capacitor starting at 15 V, each voltage falls to 36.8 % at its own time constant. The readings mirror Part II: every 1τ voltage is 15 V minus its charging value, and Circuit 3 again takes the longest and Circuit 4 the shortest.

3.1 Multisim

The Part II file was saved as a copy, the four sources were removed, and each 1 MΩ was moved across its capacitor as the discharge path (Figure 8). An initial condition on each capacitor takes the place of the charging step: 15 V on Circuits 1–3, including both capacitors in Circuit 3, since parallel capacitors share the same voltage.

Circuit 4's two capacitors are in series, so they hold the same charge, and the 15 V splits between them in inverse proportion to capacitance:

$$Q = C_{\text{EQ}} V_{\text{S}} = (31.97\ \mu\text{F})(15\ \text{V}) = 0.480\ \text{mC} \tag{8}$$

$$V_{100\ \mu\text{F}} = \frac{0.480\ \text{mC}}{100\ \mu\text{F}} = 4.80\ \text{V} \qquad V_{47\ \mu\text{F}} = \frac{0.480\ \text{mC}}{47\ \mu\text{F}} = 10.20\ \text{V} \tag{9}$$

The smaller capacitor takes the larger share, and the two add back to 15 V. The analysis was re-run to 1000 s, plotting the four voltages (Figure 9) and then the four currents (Figure 10). The probes were left as in Part II, so the current now flows out of the capacitor against the arrow and reads negative.

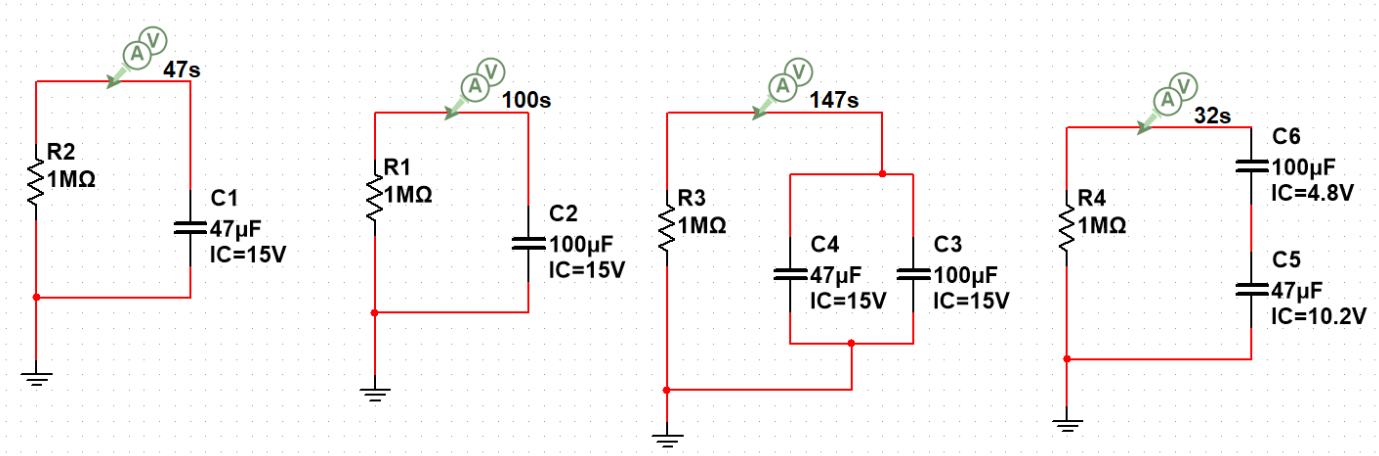


Figure 8. Part III circuits in Multisim. The sources are gone, each 1 MΩ *sits across its capacitor, and the initial conditions (IC) stand in for the charge. Circuit 4's* 4.8 V *and* 10.2 V *add to* 15 V.

3.2 Results

Circuit	τ (s)	V_C at 1τ (V)	Error (%)	I_C at 1τ (µA)	Error (%)	V_C at 5τ (V)
1	47	5.498	−0.39	−5.498	−0.39	0.095
2	100	5.526	+0.12	−5.526	+0.12	0.100
3	147	5.514	−0.11	−5.514	−0.11	0.101
4	31.97	5.677	+2.85	−5.677	+2.85	0.091

Table 4. Multisim values at 1τ and 5τ while discharging, from the exported data. Errors are against the calculated 5.52 V *and* 5.52 µA*, current by magnitude; all four should read about* 0.10 V *at 5τ.*

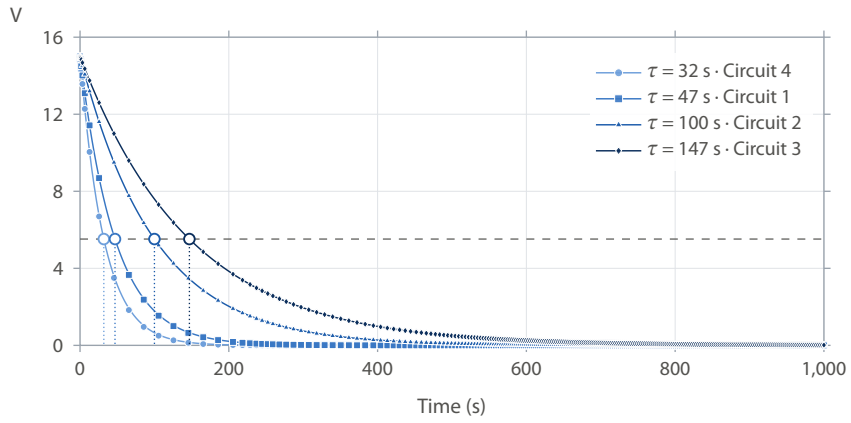


Figure 9. Capacitor voltage while discharging. Each falls to 5.52 V *at its own time constant.*

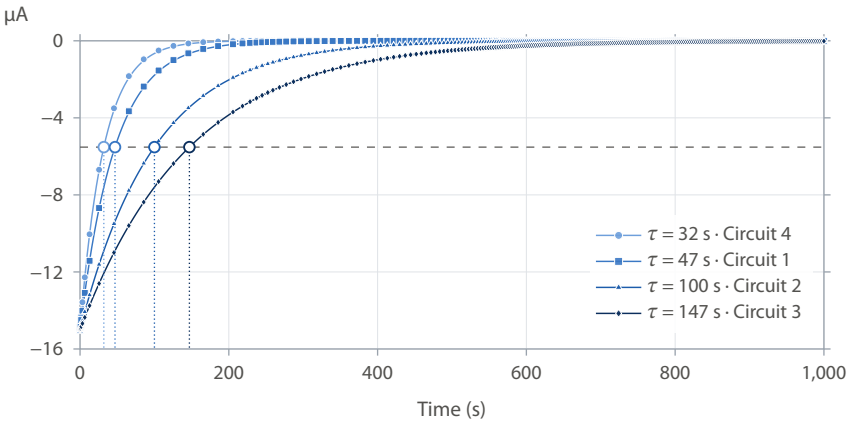


Figure 10. Capacitor current while discharging. Each starts at −15 µA*, negative because the current now leaves the capacitor against the probe arrow.*

Circuit 3 took the longest to discharge and Circuit 4 the shortest (Figure 9), the same order as charging and for the same reason. More capacitance holds more charge at 15 V, and the same 1 MΩ drains it no faster than 15 µA.

The current curves rank the same way (Figure 10), and they have to: with only the 1 MΩ in the loop, the current is the capacitor voltage divided by 1 MΩ.[3] Each current curve is its voltage curve flipped negative, which is why the readings in µA match the readings in volts in every row of Table 4.

Each 1τ reading is the Part II value turned over, $5.498\ \text{V} = 15\ \text{V} - 9.502\ \text{V}$ for Circuit 1 and likewise for the rest. Charging and discharging follow the same exponential, one rising toward 15 V and the other falling toward 0 V. Circuit 4's +2.85 % is the same 20 s point spacing as in Part II. At 5τ it reads 0.091 V against about 0.10 V, but a percent error on a value that small mostly measures the interpolation.

[3] One volt across 1 MΩ drives one microamp, so for these circuits volts and microamps are the same number.